

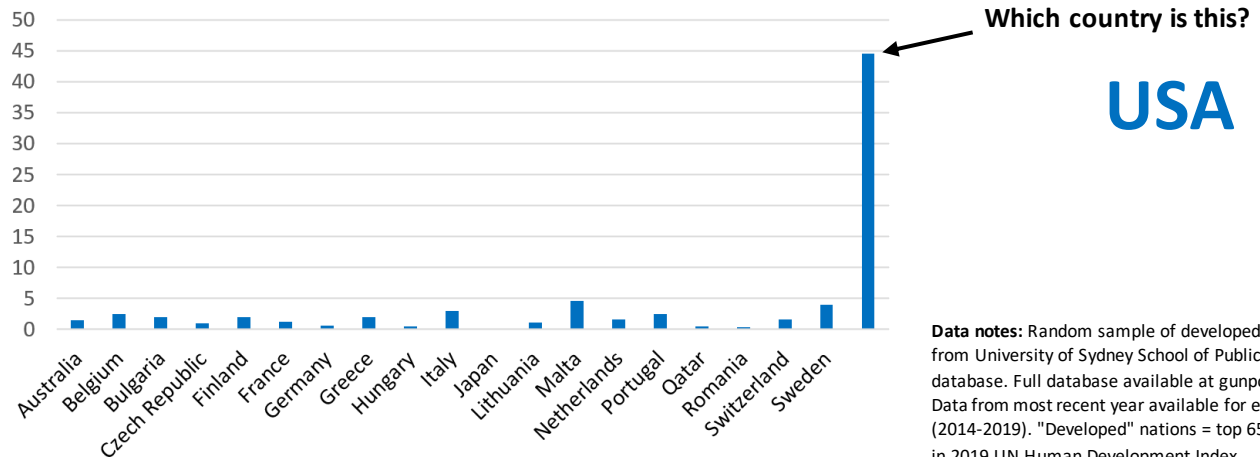
AP Statistics Handout Key: Lesson 9.3

Topics: chi-square vs. regression, t-interval for regression slope, t-test for regression slope

Lesson 9.3 Guided Notes

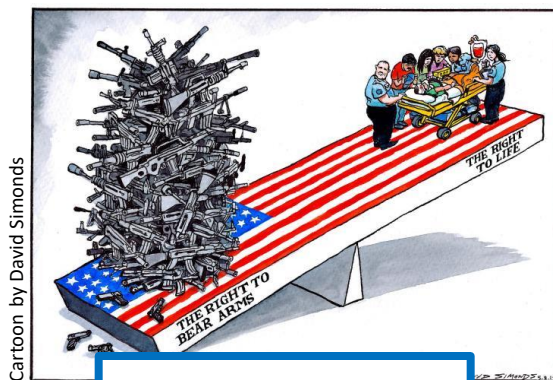
Note: Lesson inspired by "Silence the Violence" materials from 21st Century Math Projects

Gun Homicides per 1,000,000 people

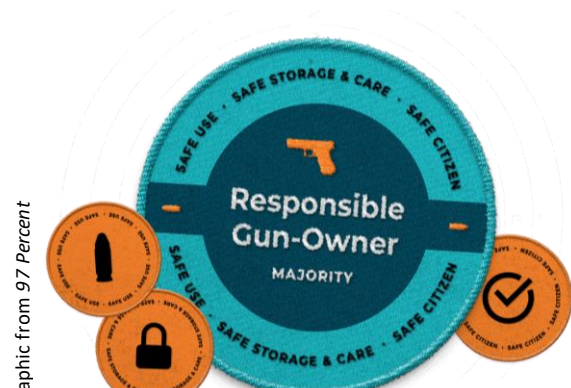


Data notes: Random sample of developed nations from University of Sydney School of Public Health database. Full database available at gunpolicy.org. Data from most recent year available for each nation (2014-2019). "Developed" nations = top 65 countries in 2019 UN Human Development Index.

Two positions for preventing gun homicide:



Gun Control



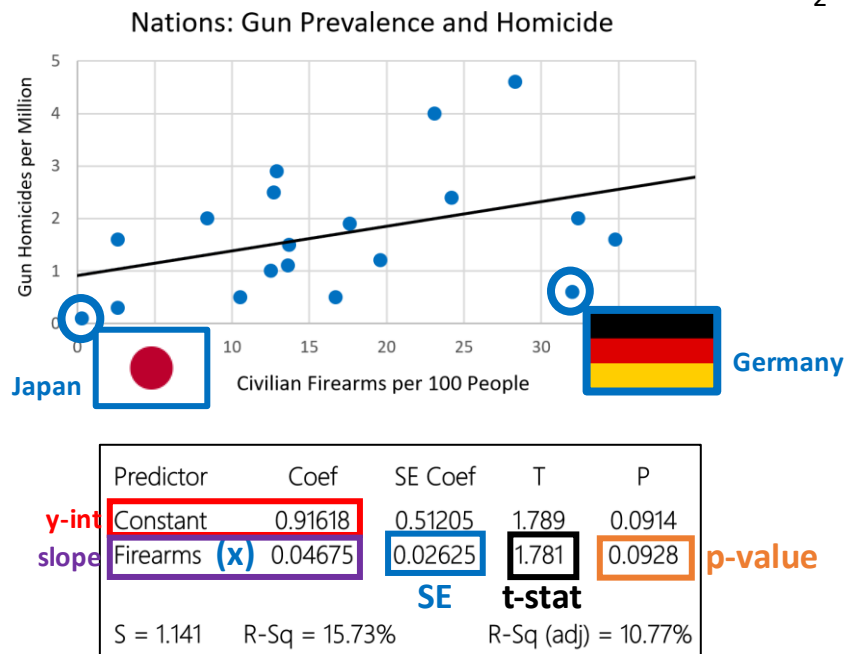
Gun Rights

Describe the arguments both of these sides make, as well as the nations they reference when making these arguments:

Gun control: Stronger gun laws would prevent guns from getting into the wrong hands and, therefore, prevent gun homicides. This side points to **Japan**, which has strict gun control and few gun homicides per year.

Gun rights: If more common citizens owned guns, it'd provide greater deterrence against criminals committing gun crimes. This side points to **Germany**, where the civilian gun ownership rate is pretty high and the gun homicide rate is fairly low.

Nation	Civilian Firearms per 100 People	Gun Homicides per 1,000,000 People
Australia	13.7	1.5
Belgium	12.7	2.5
Bulgaria	8.4	2.0
Czech Republic	12.5	1.0
Finland	32.4	2.0
France	19.6	1.2
Germany	32.0	0.6
Greece	17.6	1.9
Hungary	10.55	0.5
Italy	12.89	2.9
Japan	0.3	0.1
Lithuania	13.6	1.1
Malta	28.3	4.6
Netherlands	2.6	1.6
Portugal	24.22	2.4
Qatar	16.7	0.5
Romania	2.6	0.3
Switzerland	34.8	1.6
Sweden	23.1	4.0



Data notes: Random sample of developed nations from University of Sydney School of Public Health database. Full database available at gunpolicy.org. Data from most recent year available for each nation (2014-2019). "Developed" nations = top 65 in 2019 UNHDI.

Quick Review: Linear Regression

Data among a random sample of $n = 19$ developed nations is shown above. The data tracks each nation's firearm ownership rate (civilian firearms per 100 people) and gun homicide rate (gun homicides per million people). A linear regression line was fit for this data, with computer output shown.

1) In this case, which variable is the explanatory and which is the response? How do you know?

Civilian firearm ownership rate is the explanatory variable (x) and gun homicide rate is the response (y). We're seeing if gun ownership rates predict or explain homicide rates.

2) Circle and label Japan and Germany on these graphs. Comment on how their locations in these graphs relate to the "gun control" vs. "gun rights" arguments:

Both nations have relatively low gun homicide rates (low values on y-axis) but very different gun ownership rates (x values). Japan seems to show that low gun ownership leads to low gun homicide, whereas Germany's seems to show that high gun ownership leads to low gun homicide.

3) Comment on the overall direction and strength of the relationship:

The relationship between gun ownership and homicide rates is positive (as ownership rates increase, homicide rates tend to increase) and weak (response values tend to vary widely from LSRL predictions).

4) Write down the LSRL equation: $\hat{y} = 0.916 + 0.047x$

Where:

$\hat{y}$ = predicted gun homicides per million
 x = civilian firearms per 100 people

5) Interpret the estimate for the slope:

For every 1-unit increase in civilian firearms per 100 people, the model predicts an average increase of 0.047 gun homicides per million people.

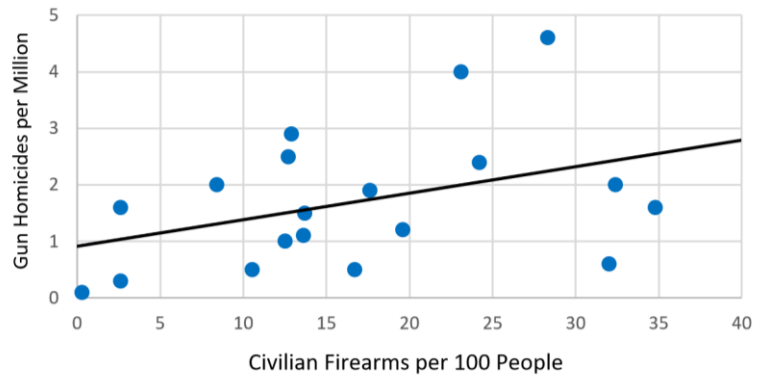


Predictor	Coef	SE Coef	T	P
Constant	0.91618	0.51205	1.789	0.0914
Firearms	0.04675	0.02625	1.781	0.0928

SE_b

S = 1.141 R-Sq = 15.73% R-Sq (adj) = 10.77%

Slope (b)



t-Interval for the Slope of a Regression Line

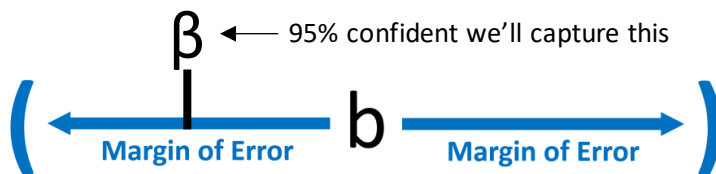
Construct and interpret a 95% confidence interval for the population regression line. First, identify the following:

Population: all developed nations

Parameter: β – true slope for regression of all developed nations

Sample: 19 sampled nations

Statistic: b – sample regression slope ($b = .047$)



Statistic $\pm$ margin of error
Statistic $\pm$ (critical value)(standard error)

Number of standard errors in either direction.

Using the formula, the computer output, and technology, **calculate and interpret** the *t*-interval for the slope of the regression line:

$$b \pm (t^*)(SE_b)$$

$$0.047 \pm (2.11)(0.026)$$

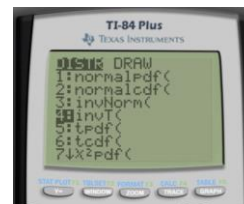
$$(-0.01, 0.10)$$

We are 95% confident the interval from -0.01 to 0.10 captures the true regression slope for the linear model relating developed nations' rates of gun ownership and gun homicide.

Confidence Interval for Regression Slope:

$$b \pm (t^*)(SE_b)$$

To find t^* with TI-84:



2nd $\rightarrow$ VARS $\rightarrow$ 4: InvT

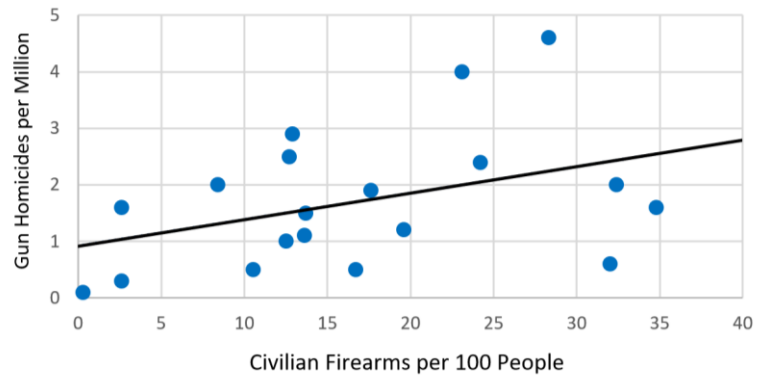


-Area: percent below interval ($2.5\% = 0.025$)
-df: $n - 2$ ($19 - 2 = 17$)

Predictor	Coef	SE Coef	T	P
Constant	0.91618	0.51205	1.789	0.0914
Firearms	0.04675	0.02625	1.781	0.0928

t-stat p-value

S = 1.141 R-Sq = 15.73% R-Sq (adj) = 10.77%



t-Test for the Slope of a Regression Line

Is there **convincing evidence** of a linear relationship between rates of gun ownership and rates of gun homicide in developed nations?

Hypotheses: $H_0: \beta = 0$ β : The **true** slope of the population regression line relating rates of gun ownership and gun homicide in **all** developed nations.
 $H_A: \beta \neq 0$

i) In your own words, interpret what the null and alternative hypotheses mean in this context:

Null: There is **no** linear relationship (neither a positive nor negative correlation) between nations' rates of gun ownership and gun homicide.

Alternative: There is a linear relationship (either a positive nor negative correlation) between nations' rates of gun ownership and gun homicide.

ii) Why do our hypotheses use β instead of b ?

We are hypothesizing about the population slope β relating the gun ownership and homicide rates among *all* developed nations. There is no need to hypothesize about b (the sample slope), since we already know the slope among our sample. We use b to make inferences about the unmeasured β .

iii) Why do you think the above test is the same thing as testing if there is **no correlation** ($\rho = 0$) between rates of gun ownership and homicide?

If there is no correlation, the scatterplot would look completely random, without any shape. For such a relationship, the slope of a LSRL would be zero.

Conducting the test:

iv) You can use the interval we calculated earlier to figure out the result of this significance test. Please do so and explain your results:

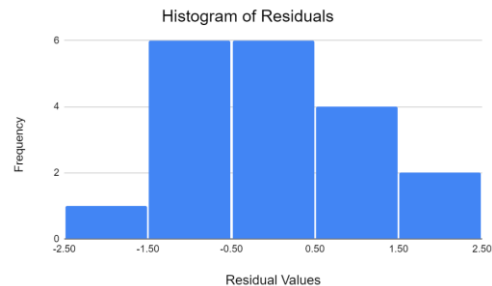
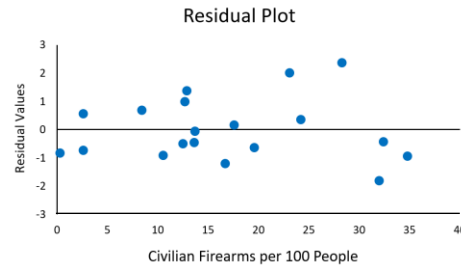
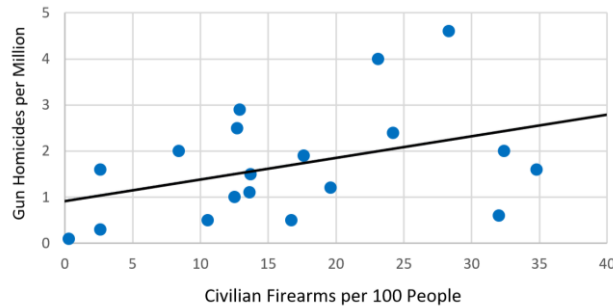
Confidence interval: **(-0.01, 0.10)**. The null value is **inside our interval** (the interval includes zero), so we cannot reject H_0 ! According to our interval, 0 is a plausible value for β . In other words, it's plausible that **there is no linear relationship** between rates of gun ownership and gun homicide.

v) Report the p-value and test statistic. Circle where they came from in the computer output above. Then, write the conclusion of your test:

t-test statistic: 1.781, p-value: 0.0928. Because our p-value (0.0928) is greater than our α level (0.05), we fail to reject H_0 . We do not have convincing evidence of a linear relationship between developed nations' rates of gun ownership and gun homicide.

Four Step Process (for both interval and test)

Rates of gun ownership and rates of gun homicide from 19 randomly sampled developed nations is shown in the scatterplot below. LSRL computer output, a residual plot, and a histogram of the residuals are provided:



Predictor	Coef	SE Coef	T	P
Constant	0.91618	0.51205	1.789	0.0914
Firearms	0.04675	0.02625	1.781	0.0928

S = 1.141 R-Sq = 15.73% R-Sq (adj) = 10.77%

Interval

Construct and interpret a 95% confidence interval for the population regression line.

State:

We are estimating β at **95% confidence**.

Where: β = The **true slope** of the population regression line relating rates of gun ownership (per 100 people) and rates of gun homicide (per 1 million people) in developed nations.

Significance Test

Is there **convincing evidence** of a linear relationship?

State:

$$H_0: \beta = 0 \quad \alpha = 0.05$$

$$H_A: \beta \neq 0$$

Where: β = The **true slope** of the population regression line relating rates of gun ownership (per 100 people) and rates of gun homicide (per 1 million people) in developed nations.

Plan: (notes on conditions included in next page, check conditions for this problem here):

Linear: The residual plot demonstrates **no leftover curved pattern**. This suggests a linear form. ✓

Independent: $n \leq \frac{1}{10} N$

$$19 \leq \frac{1}{10} (\text{all developed nations})$$

Although there are over 190 nations in the world, there are not over 190 developed nations. This condition is broken, so **we will proceed with caution!** ✗

Normal: There are no outliers and no **strong** skewness in histogram of residuals ✓

Equal SD: Residual plot shows similar scatter, with only slightly more variation at higher ownership rates. ✓

Random: the 19 countries in the dataset were randomly sampled ✓

Conditions for Regression Inference

L - Linear, to check this condition: confirm the residual plot shows no curvature in the residuals.

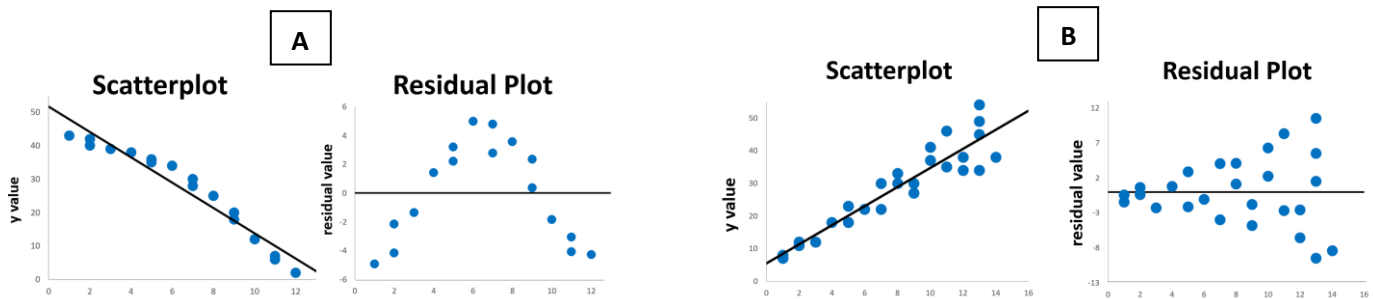
I - Independent, to check this condition: confirm that each data point is independent (knowing the value of one response value doesn't provide information about another) **OR** $n \leq \frac{1}{10}N$

N - Normal, to check this condition: confirm the distribution of residuals (in histogram) has no major outliers / no strong skewness **OR** $n \geq 30$

E - Equal SD, to check this condition: confirm the residual plot shows little **heteroscedasticity**.

R - Random, to check this condition: confirm the data comes from a random sample or a randomized experiment.

*Plots A & B are made from mock data. If you were performing inference on their datasets, these plots would be a **bad sign** for your conditions. Which conditions would they violate and why?*



A: This plot shows a violation of the linear condition, as there is a curved pattern in our residuals that is uncaptured by our linear model.

B: This plot shows a violation of the equal SD condition, as there is heteroskedasticity. The variation in our residuals changes for different x values (in this case, the variation increases as x increases)

Continuing the four steps from earlier...

Interval

Do: $\text{InvT}(\text{area: } 0.025, \text{df: } 17) \rightarrow t^* = 2.11$

$$b \pm (t^*)(SE_b) \quad \text{Interval:} \\ 0.047 \pm (2.11)(0.026) \quad (-0.01, 0.10)$$

Conclude:

We are 95% confident the interval from -0.01 to 0.10 captures the true regression slope for the linear model relating developed nations' rates of gun ownership and gun homicide.

Significance Test

Do:

t-test statistic: 1.781 df = 17

p-value: 0.0928.

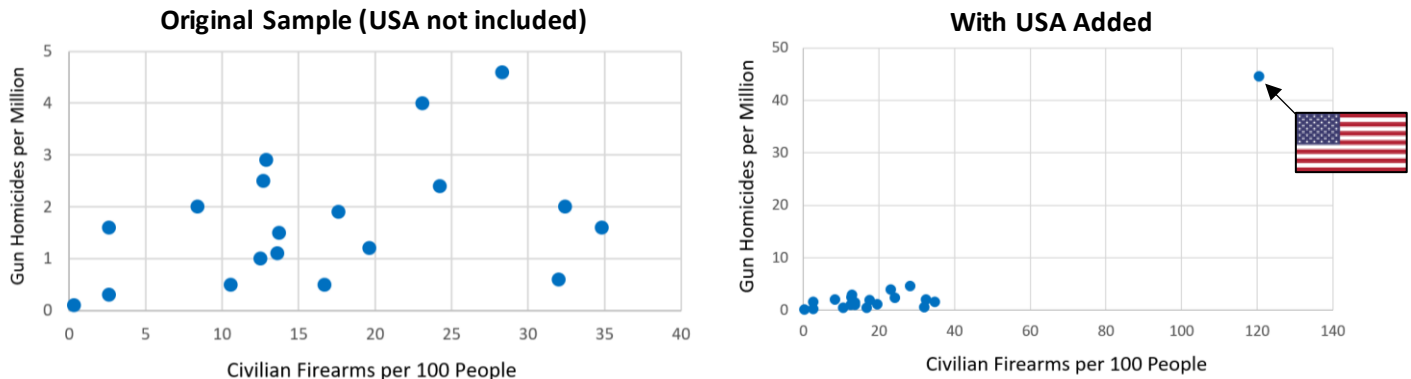
Conclude: Because our p-value (0.0928) is greater than our α level (0.05), we fail to reject H_0 . We do not have convincing evidence of a linear relationship between developed nations' rates of gun ownership and gun homicide.



Lesson 9.3 Discussion

Recommended discussion norms:
skewthescript.org/discussion-norms

We **did not find** convincing evidence of a relationship between rates of gun ownership and rates of gun homicide. This conclusion makes sense: the relationship was **very weak**.



Discussion Question: Would reducing gun ownership also reduce gun homicides in the U.S.? Explain.

- **First:** Answer this question using only the data we've analyzed, without making any assumptions.
- **Second:** Answer by choosing a clear side and use a BS (Bad Statistics) argument to support it.
- **Third:** Provide your actual personal opinion. Make sure, as always, to address the data.

First: Because the United States has an unusually high gun ownership rate compared to the countries studied, it'd be difficult to apply our analysis to the United States. In addition, we didn't find a significant relationship *among* the countries we studied. So, if we make no assumptions, we cannot answer the question one way or another.

Second: Answers will vary. Some common "bad" data arguments:

- Using the results from just one country (e.g. Japan/Germany) to make inferences about the U.S.
- Saying the *positive relationship* provides enough justification to infer that reducing gun ownership would definitely reduce gun homicides in the U.S. (ignores that the relationship was insignificant, is not necessarily generalizable to the U.S., and isn't necessarily causal)

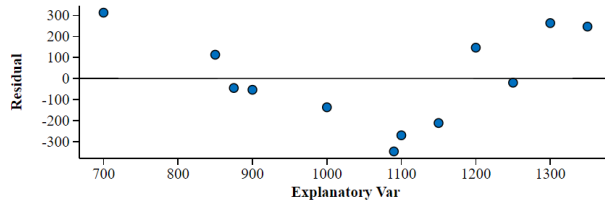
Third: Answers will (again) vary. Some important notes:

- There are many factors that prevent us from being able to successfully apply our linear regression analysis to the United States. Among these: The relationship we found was insignificant, the model is not necessarily generalizable to the U.S., and the model merely provides a correlative analysis (rather than causal).
- There is a vast network of underlying factors (differences in culture, laws, geography, socioeconomic factors, etc.) that may make the relationship between gun ownership and gun homicides different in the U.S. than in other developed nations. One could make arguments as to whether these differences would make gun control work well or work poorly in the U.S. However, it's impossible to make these arguments just using the data from this lesson.
- Gun homicides are one way to study gun violence. But fatalities from gun accidents, gun suicides, and mass shootings are also worth studying and may lead to different conclusions.

Lesson 9.3 Practice

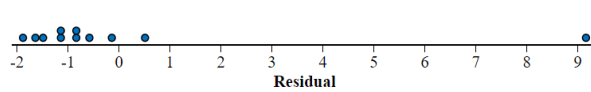
1) State which condition for linear regression inference is not met based on the given information. Then, describe what would be needed for the condition to be met.

a) Residual plot:



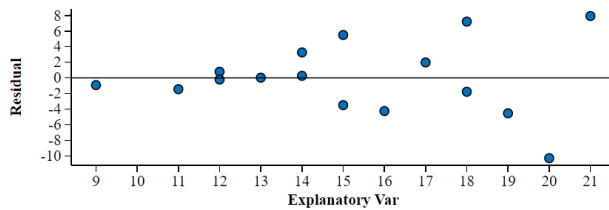
The linear condition is not met because there is a curved pattern in this residual plot. For the condition to be met, we should see random scatter with no curve.

b) Dot plot of the residuals:



The normal condition is not met because the dot plot of the residual shows an apparent outlier at approximately 9. For the condition to be met, this graph needs to lack strong skewness and outliers.

c) Residual plot:



The equal SD condition is not met because the variation in the residuals increases as the explanatory variable increases. For the condition to be met, the variability must be approximately equal across all values of x .

2) A random sample of 17 high school students from a large high school were asked how many AP courses they are taking this year and how many extracurricular activities they participate in. Computer output for the linear regression is provided below for # of extracurricular activities vs. # of AP courses.

Assume the conditions for inference have been met.

Predictor	Coef	SE Coef	T	P
Constant	1.252	0.407	3.073	0.008
# of AP courses	0.528	0.131	4.039	0.001
$s = 0.981$		$R\text{-Sq} = 52.1\%$		$R\text{-Sq (adj)} = 51.9\%$

a) We are interested in estimating the slope of the true population regression line with a confidence interval. What is the value of the point estimate for the interval?

Sample slope = $b = 0.528$

b) Define the parameter we will be estimating.

β = the true slope of the population regression line relating the number of AP courses and the number of extracurricular activities for all students at this high school.

- c) For a 95% level of confidence, what critical value would be used?

$$df = 17 - 2 = 15 \quad \text{InvT}(\text{area: } 0.025, df: 15) \rightarrow t^* = 2.131$$

- d) Calculate the margin of error for the 95% confidence interval.

$$\text{Margin of error} = t^*(SE_b) = 2.131(0.131) = 0.279$$

- e) Calculate and interpret the 95% confidence interval.

$$0.528 \pm 0.279 = (0.249, 0.807)$$

We are 95% confident the interval between 0.249 and 0.807 captures the true slope of the population regression line relating the number of AP courses and the number of extracurricular activities for all students at this high school.

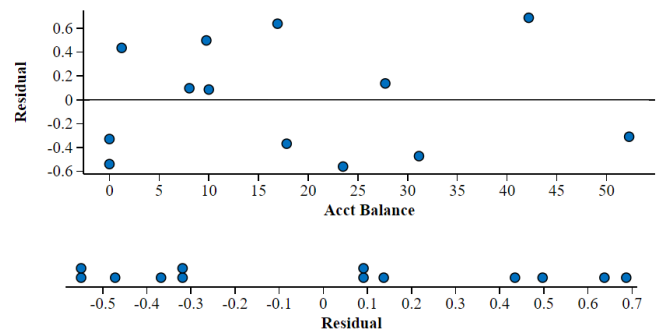
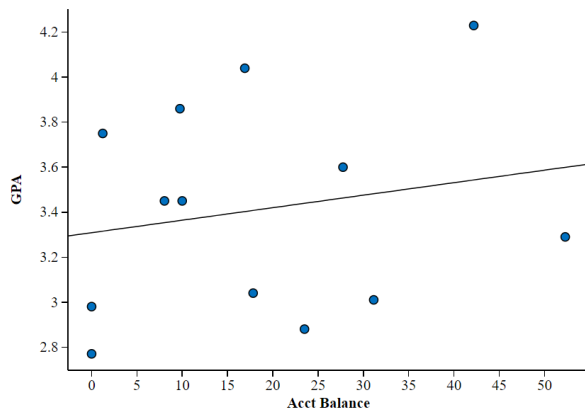
- f) Would it be appropriate to claim that there is a positive linear relationship between the number of AP courses and number of extracurricular activities for the population? Explain.

Yes. The 95% confidence interval does not contain 0 and only contains positive values as being plausible for the true slope that related number of AP courses and number of extracurricular activities.

3) Does a student's GPA relate to their lunch account balance? A random sample of 13 high school students was selected from a large high school. Students at this school may add money to a lunch account, and the account balance at the half-way point of the year was recorded for each student along with their GPA. LSRL computer output, a scatterplot, a residual plot, and a dot plot of the residuals are provided below for this data set.

Predictor	Coef	SE Coef	T	P
Constant	3.309	0.205	16.159	<0.001
GPA	0.006	0.008	0.66	0.523

s = 0.478	R-Sq = 3.8%	R-Sq (adj) = 3.4%
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Construct and interpret a 90% confidence interval for the population regression line.

State: We are estimating with 90% confidence β = the true slope of the population regression line that relates lunch account balance to GPA for all students at this high school.

Plan: We will construct a t-interval for slope, if conditions are met.

Linear: There is no curved pattern in the residual plot. ✓

Independent: Assume $13 \leq 0.10$ (all students at large high school) ← Assume true ✓

Normal: The dot plot of the residuals lacks strong skewness and outliers. ✓

Equal SD: There is roughly equal variability in the residuals across the x values. ✓

Random: The students were selected randomly. ✓

Do: $df = 13 - 2 = 11$

$$\begin{aligned} & b \pm t^*(SE_b) \\ & 0.006 \pm 1.796(0.008) \\ & = (-0.01, 0.02) \end{aligned}$$

Conclude: We are 90% confident the interval between -0.01 and 0.02 captures the true slope of the population regression line that relates lunch account balance to GPA for all students at this high school.

4) The CEO of a new tech start-up is hoping to build a team of employees that will stay with the company for a long tenure. She wants to explore the relationship between salary and tenure of employees. She collects a random sample of 18 tech start-ups and collects data on their median salary and median tenure (in years) among employees. **Assume the conditions for inference were met.** The following gives the LSRL computer output for the data:

Predictor	Coef	SE Coef	T	P
Constant	-1.099	1.718	-0.64	0.531
Median Salary	0.055	0.02	2.755	0.014
s = 1.309		R-Sq = 32.2%		R-Sq (adj) = 31.6%

- a) The CEO wants to know if there is a significant linear relationship between x = median salary (\$) and y = median tenure (years) and will perform a t-test for slope with a 5% level of significance. Write the hypotheses for this test and define the parameter you use.

$$H_0: \beta = 0$$

$H_A: \beta \neq 0$, where β = the true slope of the population regression line that relates median salary to median tenure for tech start-ups.

- b) What is the value of the test statistic?

$$t = 2.755$$

- c) What is the p-value for this test? Interpret this value in context.

The p-value is 0.014. In a world where the true slope of the population regression line is 0, there is a 1.4% chance of observing a sample slope with a magnitude of at least 0.055 by chance alone.

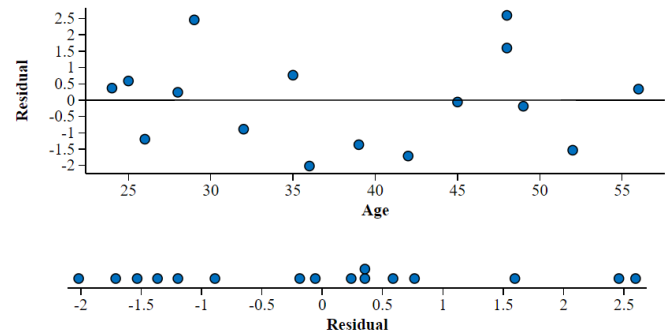
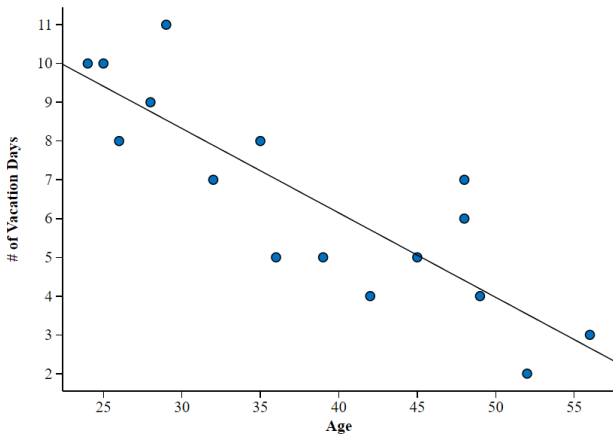
- d) Suppose the CEO believed there would be a positive linear relationship between x = median salary (\$) and y = median tenure (years). She wanted to perform the t-test on the same data to test for a significant positive linear relationship. What would be the p-value for this test?

Because the sample slope was positive (matching the direction of H_A), the p-value for the one-sided test is simply half the p-value for the two-sided test: $\frac{0.014}{2} = \mathbf{0.007}$

5) A random sample of 16 full-time workers in the U.S. were surveyed and asked their age as well as how many vacation days they used last year. LSRL computer output, a scatterplot, a residual plot, and a dot plot of the residuals are provided below.

Predictor	Coef	SE Coef	T	P
Constant	14.858	1.434	10.363	<0.001
Age	-0.218	0.036	-6.029	<0.001

$s = 1.465$	$R\text{-Sq} = 72.2\%$	$R\text{-Sq (adj)} = 70.9\%$
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Is there convincing statistical evidence of a linear relationship between age and vacation days used?

State: $H_0: \beta = 0$

$H_A: \beta \neq 0$, where β = the true slope of the population regression line that relates age to number of vacation days used for U.S. full-time workers.

$\alpha = 0.05$

Plan: We will construct a t-interval for slope, if conditions are met.

Linear: There is no curved pattern in the residual plot. ✓

Independent: $16 \leq 0.10$ (all full-time workers in the U.S.) ← Clearly true ✓

Normal: The dot plot of the residuals lacks strong skewness and outliers. ✓

Equal SD: There is roughly equal variability in the residuals across all values of age. ✓

Random: The workers were selected randomly. ✓

Do: $df = 14$ $t = -6.029$ $p\text{-value} < 0.001$

Conclude: Because the p-value (<0.001) is less than our alpha level (0.05), we reject H_0 . We have convincing evidence of a linear relationship between age and number of vacation days used for all U.S. full-time workers.

Further Practice

Teachers: If you'd like to give students additional practice problems...

- Check out our [CED and Textbook Alignment Guide](https://skewthescript.org/ap-stats-alignment) (skewthescript.org/ap-stats-alignment) to find additional exercises in your AP Stats textbook or in AP Classroom that are aligned to the content covered in this lesson.
- The following Free Response Questions (FRQs) from [past AP Exams](#) are also aligned to this lesson: 2011 Q5 (part d), 2008 Q6 (part c)

